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PROJECT REPORT

AyuSetu: Integrated Telemedicine & Panchakarma Care Platform

Project Title: AyuSetu

Tagline: Bridging Traditional Healing and Modern Healthcare for Everyone

Domain: Healthcare / Telemedicine / Ayurveda

1. Introduction

1.1 Objective of the Project

The primary objective of **AyuSetu** is to create a unified digital ecosystem that bridges the gap between modern medical access in rural India and traditional Ayurvedic therapy management. The core goals are:

- **Democratize Healthcare Access:**
  - To provide rural patients with immediate access to qualified specialists via video/audio consultations.
  - To reduce the financial burden and time lost in traveling long distances for basic medical advice.
- **Modernize Ayurveda:**
  - To digitize the currently unstructured management of Panchakarma therapies.
  - To move from manual paper records to automated scheduling, tracking, and patient notifications.
- **Unified Care Model:**
  - To bring Allopathic (Modern) and Ayurvedic (Traditional) treatments under a single "Phygital" (Physical + Digital) roof.
  - To allow patients to switch seamlessly between consultation types based on their needs.
- **Empower Stakeholders:**
  - To enable doctors to manage remote practices efficiently.
  - To help local pharmacies increase visibility by displaying real-time medicine stock to nearby patients.

1.2 Description of the Project

AyuSetu is a web-based platform designed to address specific challenges in India's healthcare sector. It is not just a booking app; it is a complete management system.

- **For Patients:** It acts as a digital health companion. A patient in a village can consult a city doctor, book a 7-day Ayurvedic detox therapy, and check if their prescribed medicine is available at the shop next door—all from one phone.
- **For Practitioners:** It serves as a practice management tool. Ayurvedic Vaidyas can manage complex, multi-day therapy schedules without overlapping appointments.
- **For Infrastructure:** It includes an offline-first architecture. Recognizing that rural internet is unstable, the system allows key data (like health records) to be accessed without an active connection.
- **AI Assistance:** An integrated AI symptom checker helps patients triage their own condition (e.g., "Is this fever urgent?") using simple voice or text inputs.

1.3 Scope of the Project

The scope defines the specific boundaries of what the AyuSetu system will cover.

In-Scope:

- **User Modules:**
  - **Patient Module:** Registration, Search, Booking, History.
  - **Doctor Module:** Calendar, Video Interface, Digital Prescriptions.
  - **Pharmacy Module:** Stock updates, Order viewing.
  - **Admin Module:** User verification, Analytics.
- **Core Features:**
  - Real-time video and audio consultations (WebRTC).
  - Panchakarma Therapy Management (Scheduling logic for multi-day sessions).
  - AI-based Symptom Checker (NLP model).
  - Notification Engine (SMS/Email/Push).
  - Offline Data Synchronization.

Out-of-Scope:

- **Physical Delivery:** The system will strictly display medicine availability; it will not handle logistics or delivery boys.
- **Emergency Care:** The platform is explicitly for non-emergency consultations. Critical cases (heart attacks, trauma) will be redirected to emergency helplines.
- **Payment Gateway:** (For Phase 1) Payments will be settled offline/locally to simplify rural adoption, though the architecture will support payment integration later.

1.4 Strategic Analysis

To understand the project's viability, we conducted a SWOT analysis identifying the key strengths, weaknesses, opportunities, and threats associated with the platform.

Figure 1.1: SWOT Analysis highlighting the strategic position of AyuSetu.

1.5 Analysis of Rural vs Urban Healthcare Accessibility

A key driver for the AyuSetu project is the significant disparity between rural and urban healthcare infrastructure. Our analysis reveals critical gaps that digital intervention can address:

- **Doctor Availability:** Rural areas have almost **1 doctor per 10,000 people**, while urban regions have a significantly better ratio of **1 per 3,000**.
- **Travel Distance:** Rural patients are forced to travel **10–12 km** on average for basic treatment, compared to just **2–3 km** for city dwellers.
- **Infrastructure Quality:** Healthcare infrastructure is objectively weaker in villages, scoring approximately **3/10**, whereas urban areas average **8/10**.

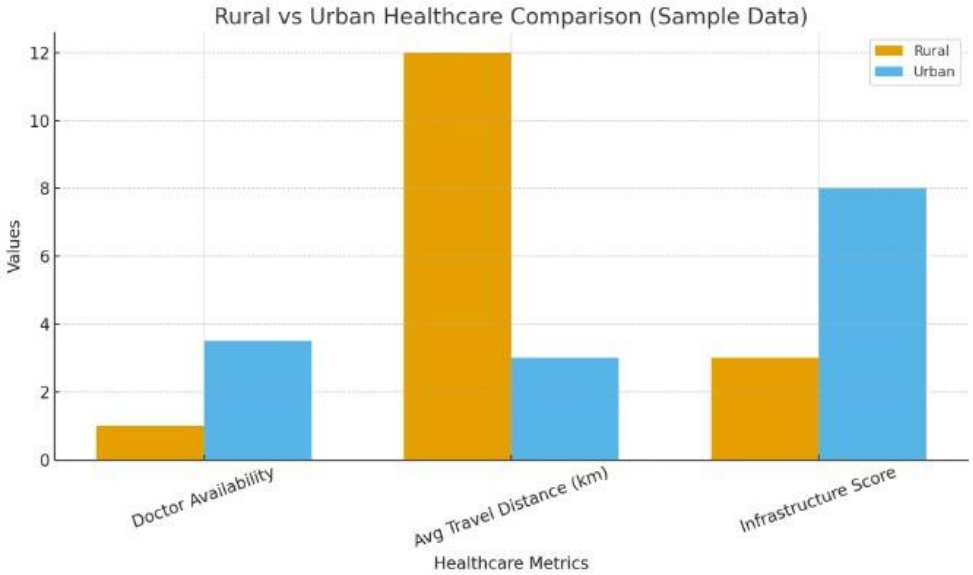

Figure 1.2: Bar chart illustrating the stark contrast in healthcare metrics between rural and urban regions, underscoring the need for telemedicine solutions.

1.6 Study on Panchakarma Therapy Utilization & Effectiveness

Our research indicates that technology plays a vital role in treatment adherence. This pie chart clearly shows how Panchakarma therapy becomes much more effective when patients receive digital support (the AyuSetu Model) compared to traditional methods without tracking.

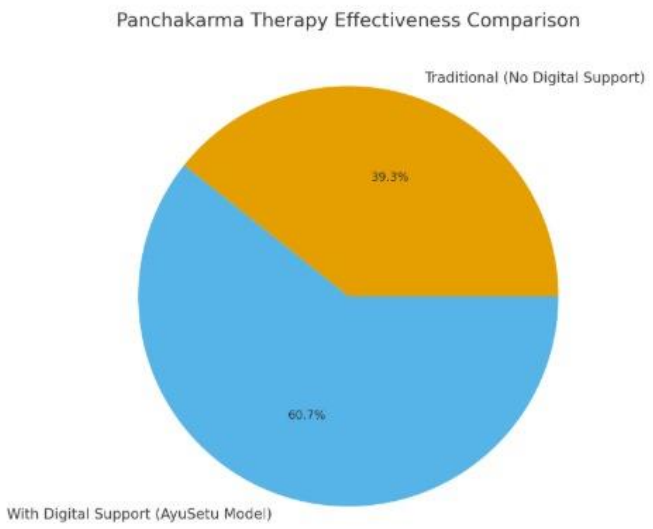

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Figure 1.3: Comparison of therapy effectiveness, showing a 60.7% effectiveness rate with digital support versus 39.3% for traditional methods.

1.7 Demographic Insights: Telemedicine vs. Ayurveda Usage

Understanding the target audience's preferences is crucial for adoption.

- **Young Adults (18–35):** Show a strong preference for **Telemedicine** (approx. 80% usage preference) due to high digital literacy and convenience.
- **Middle-aged Adults (35–55):** Drive the demand for traditional **Ayurvedic consultations** (approx. 70% preference), valuing holistic care.
- **Strategic Implication:** AyuSetu's hybrid model is essential to capture both the tech-savvy youth and the older demographics who trust traditional healing.

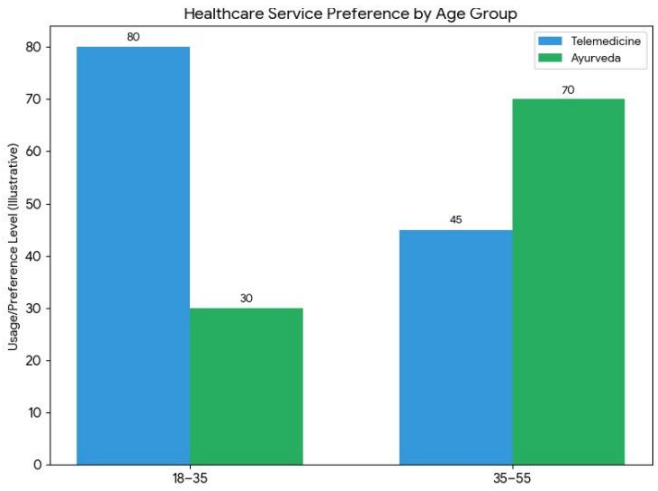


Figure 1.4: Bar chart comparing healthcare service preferences across different age groups, validating the need for a hybrid platform.

1.8 Total Healthcare Ecosystem Integration

AyuSetu integrates the entire healthcare ecosystem—including Patient, Doctor, Pharmacy, and Admin portals—into one unified platform, eliminating the need for multiple fragmented apps. This unification simplifies the user journey, ensuring that a patient can consult, book therapies, and check medicine stock without ever leaving the ecosystem.

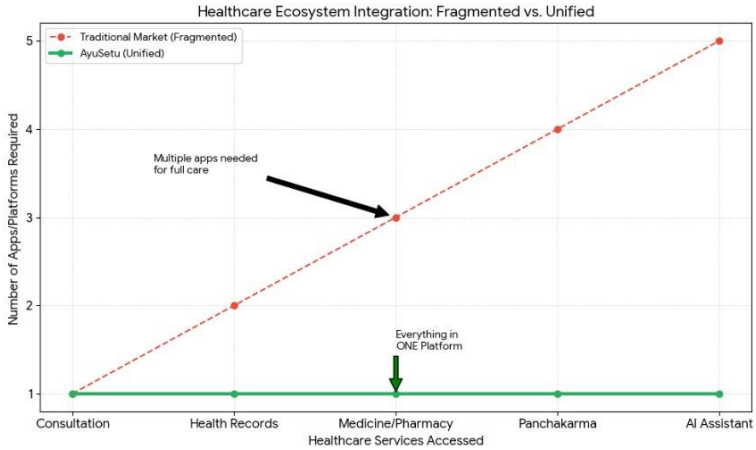


Figure 1.5: Graph illustrating the efficiency of AyuSetu's unified platform compared to the fragmented traditional market, where multiple apps are typically required for full care.

1.9 App Functionality: Cloud vs. Offline AI

Standard health apps often fail in rural settings because they continuously need to "talk" to a

server, resulting in approximately **20% functionality** in low-network zones. In contrast, AyuSetu is designed to work perfectly with **100% functionality** because the AI symptom checker and health records are engineered for offline or low-bandwidth modes.

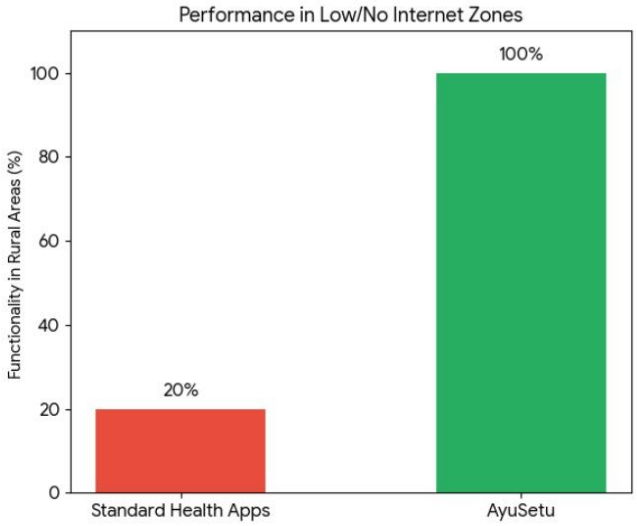


Figure 1.6: Bar chart demonstrating AyuSetu's superior performance in low/no internet zones compared to standard health apps.

2. System Description

2.1 Existing System vs. Proposed System

The Existing System (Manual Process):

- **Consultations:** Patients travel 20-50km to cities for basic checkups.
- **Records:** Paper files are often lost or damaged.
- **Ayurveda:** Panchakarma appointments are noted in diaries. If a patient forgets a session, there is no reminder system.
- **Medicines:** Patients visit multiple pharmacies physically to check for stock.

The Proposed System (AyuSetu):

- **Consultations:** Instant video connection from home.
- **Records:** Cloud-stored history accessible forever.
- **Ayurveda:** Digital calendar with automated reminders (SMS/App) for every session.
- **Medicines:** Real-time visibility of stock at local shops before leaving home.

Competitive Landscape:
While other digital platforms exist, AyuSetu distinguishes itself by focusing on rural-specific needs and integrating Ayurveda.

Competitive Analysis Table

Aspect	AYUSETU	PRACTO	LYBRATE	1MG(TATA HEALTH)
Background	Designed to bridge rural healthcare gaps and bring structured Ayurvedic and telemedicine services onto a unified digital platform.	Leading online doctor consultation and appointment booking platform.	Provides online doctor consultations with a focus on homeopathy and Ayurveda.	E-commerce and consultation platform with Ayurvedic and Allopathic medicines.
Products/Services	Teleconsultations, Panchakarma therapy scheduling, AI health guidance, health record management, pharmacy integration, and notifications/reminders.	Doctor consultations (Allopathy, some Ayurveda), diagnostics, and medicine delivery.	Homeopathy and Ayurveda-focused consultations, some Allopathy.	Medicine delivery (Allopathy, Ayurveda), doctor consultations, and lab tests.
Unique Features	AI-based symptom checker, digital Panchakarma therapy management, multilingual support, low-bandwidth telemedicine, and real-time medicine stock updates.	Extensive doctor network and hospital tie-ups.	AI-driven health advice, chatbot consultation.	E-commerce focus on medicines with expert advice.
Target Audience	Rural and urban patients, Ayurvedic practitioners, doctors, pharmacies, and healthcare administrators.	General patients needing appointments & consultations.	Ayurveda and Homeopathy enthusiasts.	Online medicine buyers and health-conscious individuals.
Supply Chain Efficiency	Efficiently connects patients, doctors, and pharmacies by syncing medicine availability, therapy schedules, and digital health records in real time.	High efficiency with multiple fulfillment centers.	Moderate efficiency with limited e-commerce.	Very high efficiency with nationwide fulfillment centers.

Figure 2.1: Competitive Analysis Table comparing AyuSetu with major market players like Practo, Lybrate, and 1MG.

2.2 Customer/User Profiles

The system is designed for four distinct user groups with specific needs:

- **Rural & Urban Patients:**
  - *Demographics:* Wide range of ages, variable tech literacy.
  - *Constraints:* Potentially slow internet (2G/3G), preference for vernacular languages (Hindi/Punjabi).
  - *Requirement:* Large buttons, voice navigation, offline mode.

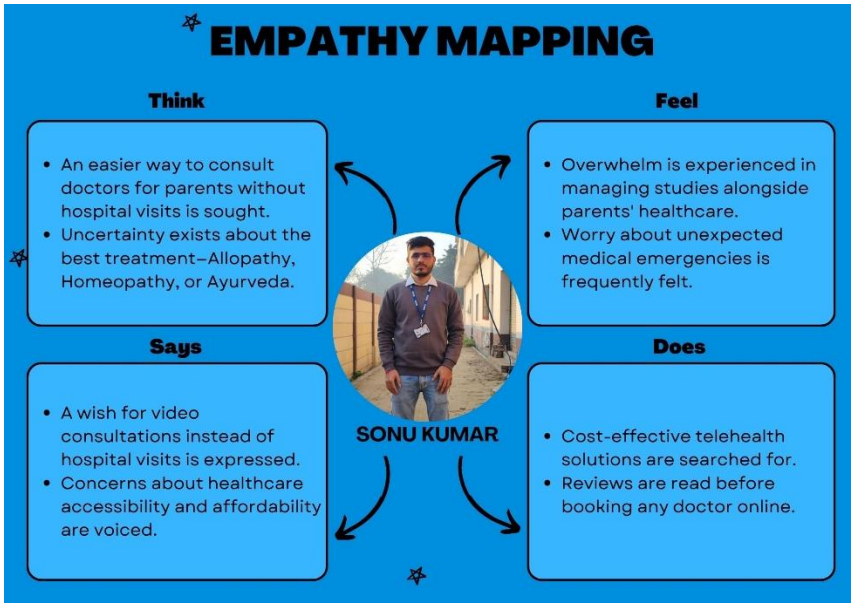


Figure 2.2: Empathy Map illustrating the typical needs ("Think", "Feel", "Say", "Do") of a student managing healthcare for their parents in a rural setting.

- **Doctors (Allopathic & Ayurvedic):**
  - *Demographics:* Certified medical professionals.
  - *Constraints:* Limited time per patient.
  - *Requirement:* Quick access to patient history, stable video quality, fast prescription typing tools.

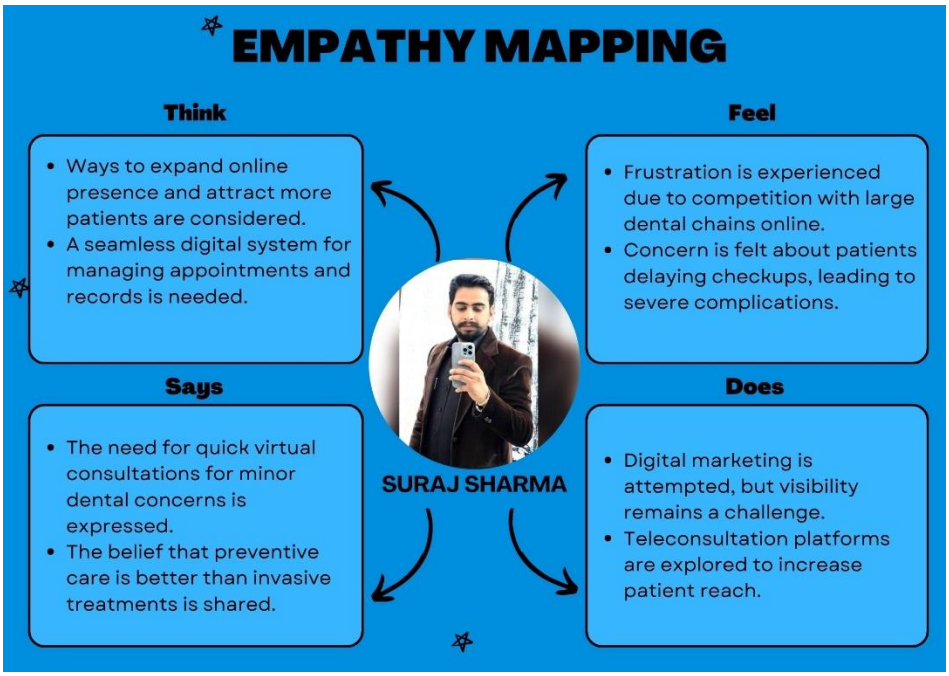


Figure 2.3: Empathy Map for a Practitioner, highlighting the need for digital tools to manage appointments and compete with larger chains.

- **Local Pharmacies:**
  - *Demographics:* Small business owners.
  - *Requirement:* Extremely simple interface to just toggle "In Stock" or "Out of Stock".
- **Administrators:**
  - *Demographics:* Health department officials or System Admins.
  - *Requirement:* High-level dashboards to monitor disease trends and system health.

2.3 Functional Requirements

These are the specific behaviors the system must support, categorized by module.

A. Patient Module

- **Registration:** Sign up using Mobile Number + OTP.
- **Profile Management:** Add age, gender, and blood group.
- **Doctor Search:** Filter by Specialty (Cardio, Ayurveda, ENT), Location, and Experience.

- **Appointment Booking:** View real-time slots and book instantly.
- **Symptom Checker:** Input symptoms (Voice/Text) -> Receive AI suggestions.

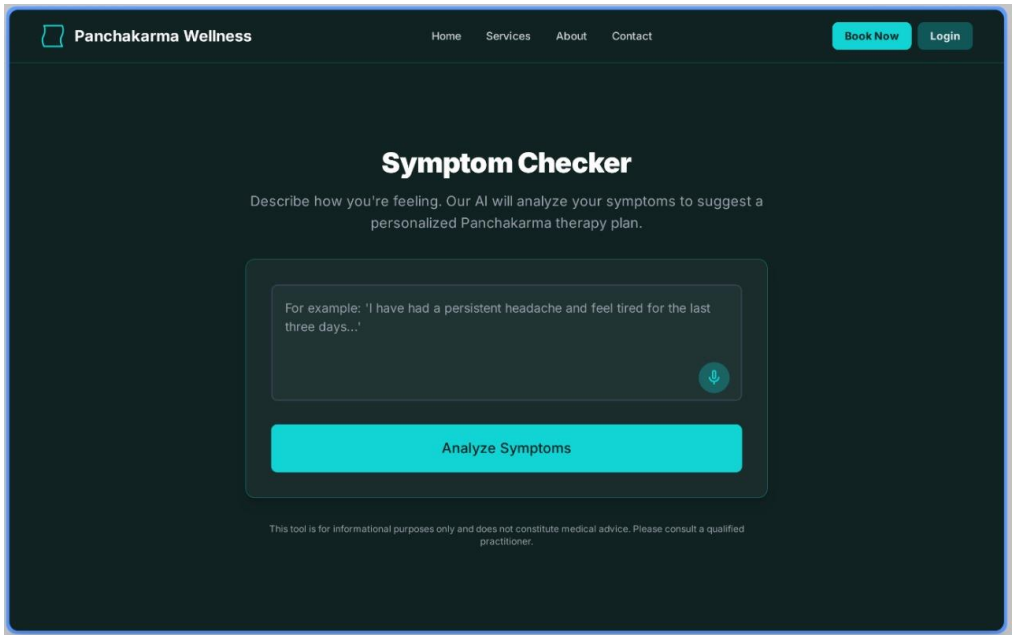


Figure 2.4: The Symptom Checker input screen allowing users to describe their condition via text or voice.

- **History:** View past prescriptions and therapy progress bars.

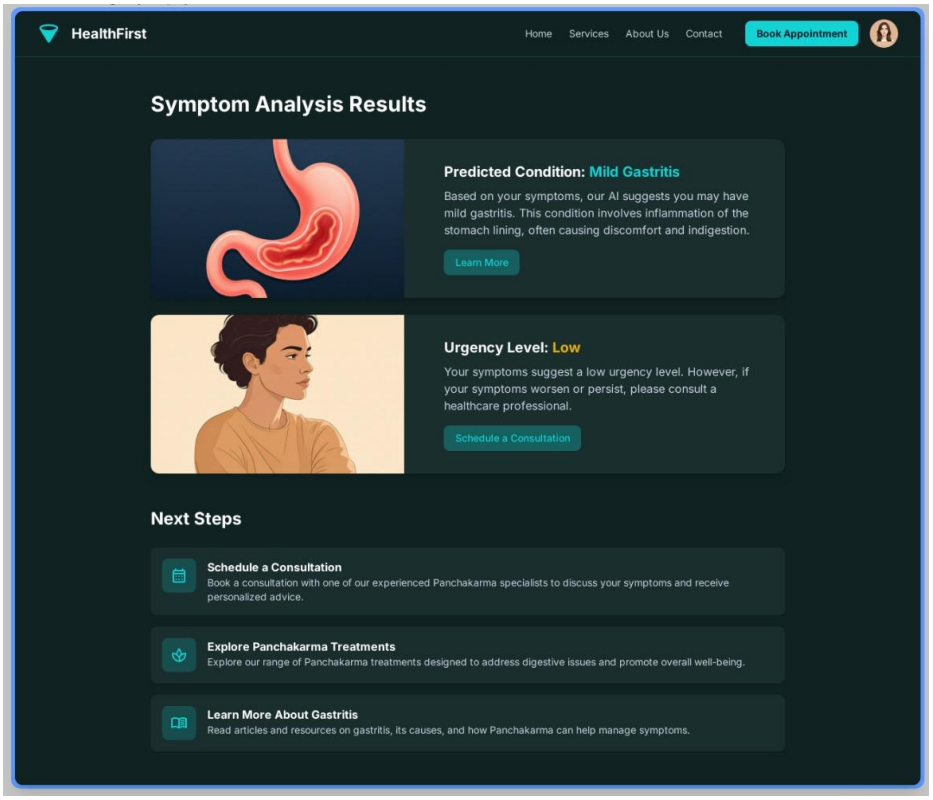


Figure 2.5: The AI-powered Symptom Checker interface displaying analysis results and recommended next steps.

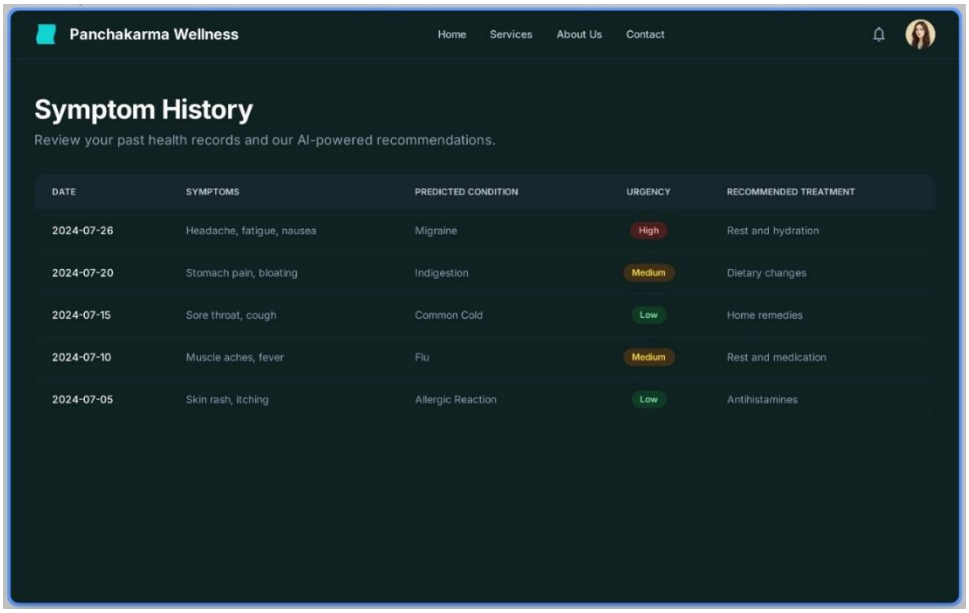


Figure 2.6: The Patient History Dashboard tracking past symptom checks, predicted conditions, and urgency levels over time.

B. Doctor Module

- **Dashboard:** View today's schedule and pending appointments.
- **Consultation Room:** Integrated video call interface with a side panel for taking notes.
- **e-Prescription:** Generate a digital PDF prescription that is instantly shared with the patient.
- **Therapy Manager:** (For Ayurvedic Doctors) Mark daily progress of Panchakarma sessions (e.g., "Day 3: Shirodhara Completed").

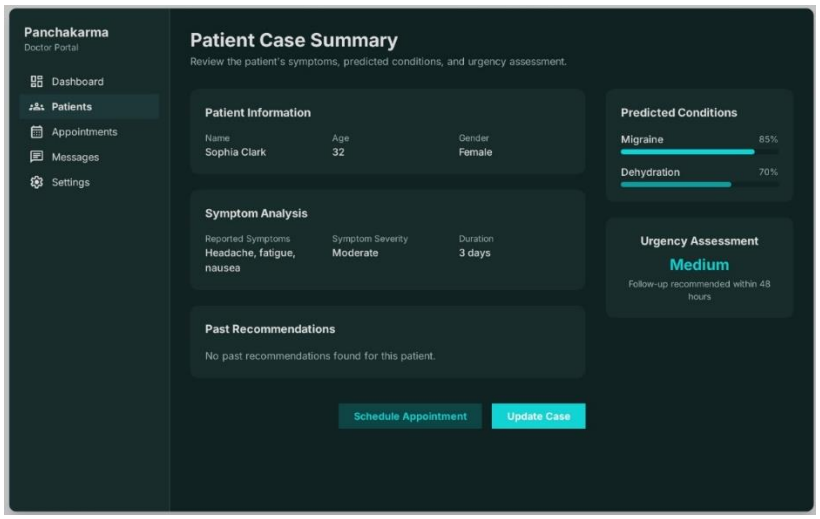


Figure 2.7: The Doctor's Case Summary view, showing consolidated patient data, AI predictions, and urgency assessment.

C. Pharmacy Module

- **Inventory Toggle:** A simple list of common medicines with "Available/Unavailable" switches.
- **Prescription View:** Scan a patient's QR code to see their valid prescription.

D. Admin Module

- **User Verification:** Approve doctor credentials before they go live.
- **Analytics:** Graphs showing:
  - Total Consultations today.
  - Top reported symptoms (e.g., "High Fever" spike in a specific district).

2.4 Non-Functional Requirements

- **Performance:** The application must load the landing page within 3 seconds on 3G networks.
- **Reliability:** The system should aim for 99% uptime, particularly for the booking engine.
- **Scalability:** The backend (Node.js) must handle up to 10,000 concurrent users as the platform expands to new districts.
- **Security:**
  - Patient health data must be encrypted at rest and in transit.
  - Video calls must use end-to-end encryption (WebRTC standard).
- **Usability:**
  - **Multilingual:** Complete UI availability in English, Hindi, and Punjabi.
  - **Accessibility:** High contrast modes and font scaling for elderly users.

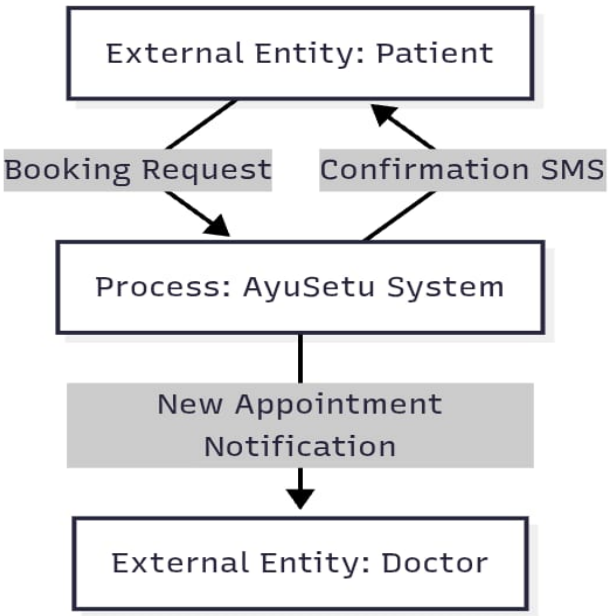
3. Design

3.1 System Design

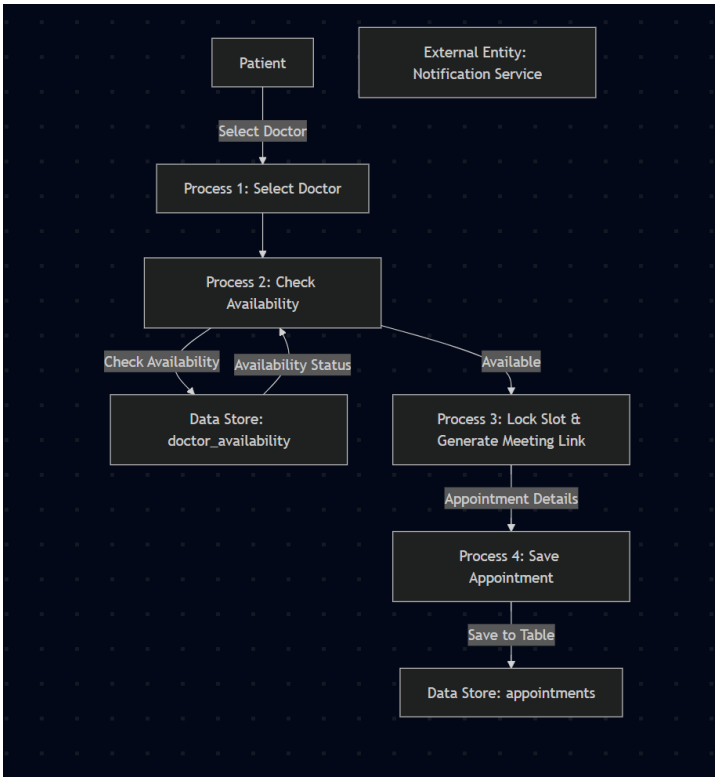
The system follows a Client-Server architecture with a Microservices approach for resource-heavy features like the AI module.

3.1.1 Data Flow Diagrams (DFD)

- **Level 0 (Context Level):**
  - **Input:** Patient sends "Booking Request".
  - **Process:** AyuSetu System checks availability.
  - **Output:** System sends "Confirmation SMS" to Patient and "New Appt Notification" to Doctor.



- **Level 1 (Consultation Process):**
  - **Step 1:** Patient Selects Doctor.
  - **Step 2:** Database checks doctor_availability table.
  - **Step 3:** If Available -> Lock Slot -> Generate meeting_link.
  - **Step 4:** Save to appointments table.
  - **Step 5:** Trigger Notification Service.



3.1.2 Information Architecture

The Information Architecture (IA) outlines the structural design of the shared information environments, showing how the application's content and navigation are organized to support usability and findability.

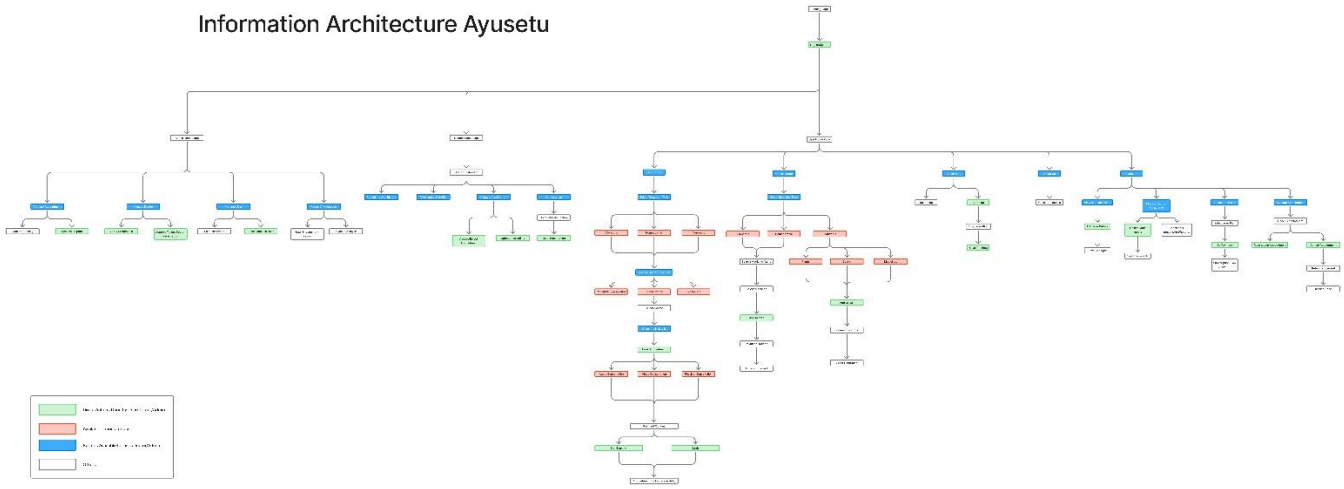


Figure 3.1: Information Architecture diagram showing the hierarchy and navigation flow of the AyuSetu platform.

3.2 User Interface Design

A key focus of the project is to provide a clean, accessible, and modern interface. The platform supports a "Dark Mode" to reduce eye strain and save battery on mobile devices in rural areas.

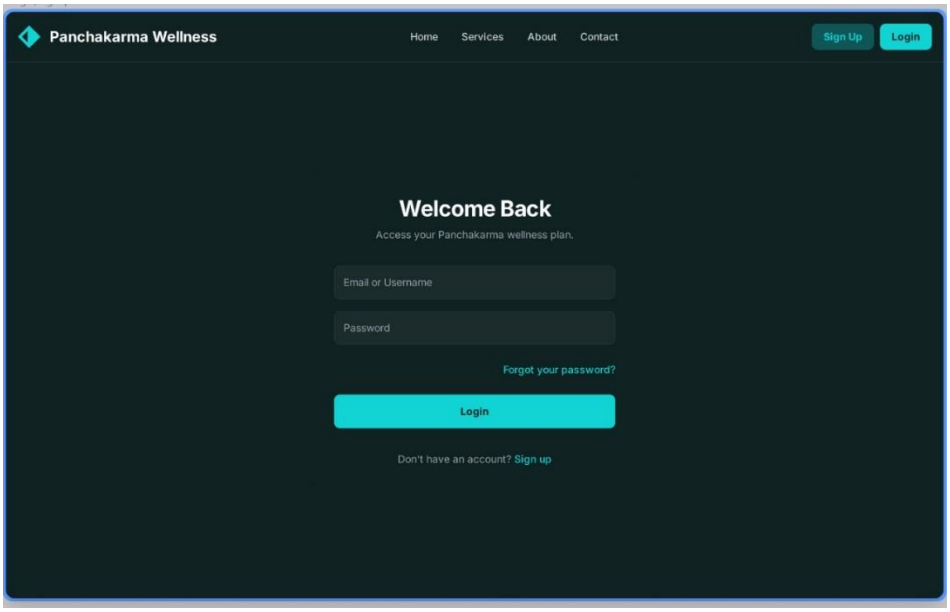


Figure 3.2: The login interface design featuring high-contrast text for accessibility and a simple, distraction-free layout.

4. Scheduling and Estimates

This section outlines the roadmap for developing AyuSetu, ensuring a structured approach from concept to launch.

4.1 Development Phases

Phase 1: Planning & Requirement Analysis (Weeks 1-2)

- **Goal:** Define exactly what to build.
- **Tasks:**
  - Interview local doctors and rural patients to understand pain points.
  - Finalize the technology stack (React, Node, PostgreSQL).
  - Define the database schema.

Phase 2: Design & Prototyping (Weeks 3-5)

- **Goal:** Visualize the system.
- **Tasks:**
  - Create UI Wireframes in Figma (Focus on mobile view).

    - Design the API structure.
    - Create the "Green & Blue" color theme (Nature + Medical).

Phase 3: Review Paper on NLP & Core Development (Weeks 6–9)

- **Goal:** Begin with the review paper on NLP and initiate core development.
- **Tasks Completed (Review Paper):**
    - Conducted a detailed review study on Natural Language Processing (NLP) focused on its applications in healthcare symptom analysis.
    - Reviewed existing NLP models, datasets, and techniques relevant to building the AI symptom checker for AyuSetu.
    - Analyzed how NLP can support low-bandwidth and multilingual environments for rural healthcare users.
- **Planned Tasks (Upcoming Weeks):**
    - **Frontend Development:** Build Patient and Doctor portals with vernacular language toggle.
    - **Backend Development:** Set up the server, authentication, and database connections.
    - **AI Integration:** Train the TensorFlow model on basic symptom datasets.

4.2 Conclusion

AyuSetu represents a pivotal step forward in harmonizing India's healthcare landscape, effectively bridging the gap between modern medical access and traditional Ayurvedic healing. By digitizing the complex workflows of Panchakarma and enabling remote teleconsultations, the platform directly addresses the "last mile" connectivity challenge in rural areas. Its robust, offline-first architecture ensures that technology serves as an enabler rather than a barrier, making quality care accessible regardless of geographical or infrastructural limitations. Ultimately, this project establishes a scalable, inclusive digital ecosystem that empowers patients, streamlines practice for doctors, and revitalizes traditional medicine for the modern age.